



Unravelling AMR dynamics in the rumenofaecobiome: Insights, challenges and implications for One Health

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ARTICLE INFO

Article history:

Received 10 July 2024

Accepted 13 March 2025

Editor: Professor Jian Li

Keywords:

Antimicrobial resistance

Rumen microbiome

Faecobiome

Livestock

Unculturable microbes

ABSTRACT

Antimicrobial resistance (AMR) is a critical global threat to human, animal and environmental health, exacerbated by horizontal gene transfer (HGT) via mobile genetic elements. This poses significant challenges that have a negative impact on the sustainability of the One Health approach, hindering its long-term viability and effectiveness in addressing the interconnectedness of global health. Recent studies on livestock animals, specifically ruminants, indicate that culturable ruminal bacteria harbour AMR genes with the potential for HGT. However, these studies have focused predominantly on using the faecobiome as a proxy to the rumen microbiome or using easily isolated and culturable bacteria, overlooking the unculturable population. These unculturable microbial groups could have a profound influence on the rumen resistome and AMR dynamics within livestock ecosystems, potentially holding critical insights for advanced understanding of AMR in One Health. In order to address this gap, this review of current research on the burden of AMR in livestock was undertaken, and it is proposed that combined study of the rumen microbiome and faecobiome, termed the 'rumenofaecobiome', should be performed to enhance understanding of the risks of AMR in ruminant livestock. This review discusses the complexities of the rumen microbiome and the risks of AMR transmission in this microbiome in a One Health context. AMR transmission dynamics and methodologies for assessing the risks of AMR in livestock are summarized, and future considerations for researching the impact of AMR in the rumen microbiome and the implications within the One Health framework are discussed.

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1. Overview of AMR as a risk to One Health

1.1. AMR is threatening One Health

According to the World Health Organization, antimicrobial resistance (AMR) is a global public health threat that directly impacts human, animal, plant and environmental health [1,2]. It carries a heavy economic burden due to high costs of treatments and reduced productivity caused by infections with resistant pathogens [3]. The US Centers for Disease Control and Prevention reports more than 2.8 million AMR infections and 1.27 million deaths worldwide each year, which is more than the number of deaths caused by human immunodeficiency virus, acquired immunodeficiency syndrome or malaria [4]. In the UK, more than 160,000

patients were identified with antibacterial or antifungal resistance infections in the bloodstream in 2022, according to records from laboratories in England [5]. AMR bacteria are commonly found between environments [6,7], posing a threat to the health of humans, animals, plants and the environment [8]. Thus, controlling the spread of AMR is a complex and challenging ecological issue [9]. Significant strides have been made in understanding the resistomes of livestock [10–14]; for instance, Pal et al. demonstrated that interconnectedness of the entire collection of resistance genes varied across human–animal–environmental microbial communities, and showed that there was no significant difference in microbiome taxa and AMR gene (ARG) diversity among these environments [15]. This interconnectedness provides the context for One Health, an interdisciplinary and integrated global framework through which the microbiologically mediated interactions among humans, animals and the environment can be investigated to promote optimal health outcomes on a global scale [9]. Climate change [16], environmental pollution [17] and the misuse of drugs

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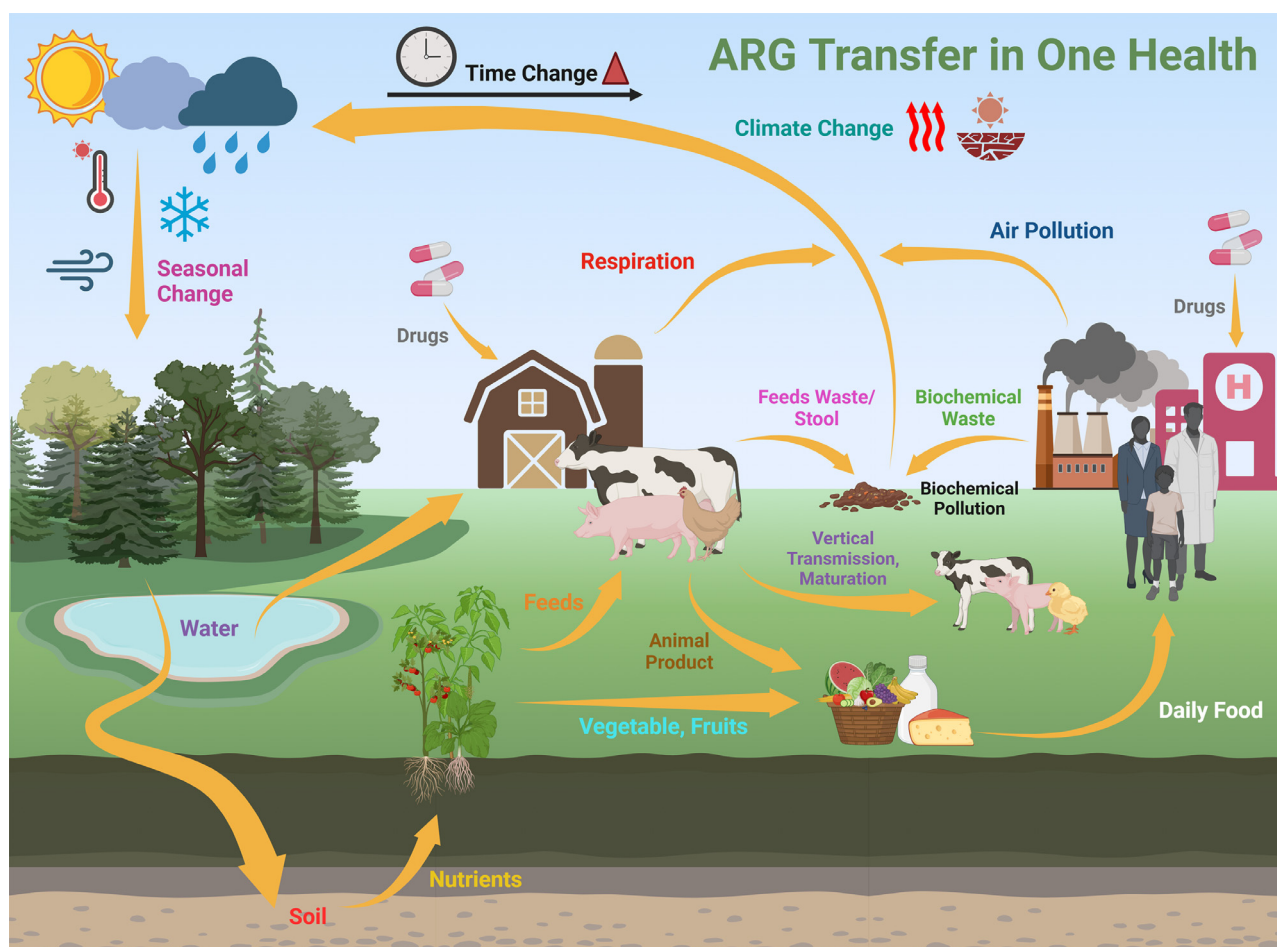


Fig. 1. Spatio-temporal representation of antimicrobial resistance gene (ARG) transfer pathways across diverse environments in One Health. The dissemination of antimicrobial resistance (AMR) operates through a continuous environmental cycle, wherein AMR determinants are absorbed by soil systems and subsequently distributed to aquatic environments and various plant species. These resistance determinants then propagate through the food chain, ultimately reaching animal and human populations. The cycle completes as AMR determinants from food/feed waste, faecal matter and respiratory emissions from humans and animals are reintroduced into the atmosphere. This cyclical process underscores the profound interconnectedness of environmental components, and emphasizes the critical One Health implications associated with AMR transmission across ecosystem. This figure was created using BioRender.

[18] lead to increased selective pressure on micro-organisms, contributing to the emergence of AMR, posing a risk to One Health (Fig. 1). Therefore, surveillance of antimicrobial utilization and the ARGs detected in humans, animals and the environment encompassing the One Health approach is crucial to address the issue of AMR [6,8,19].

Antimicrobials are indispensable chemicals in medicine and agriculture. Approximately 73% of the antimicrobials in the world are used for agricultural purposes, such as prophylactics, therapy, eradication and growth promotion [14,20–22]. The indiscriminate use of antimicrobials in agriculture, especially in livestock production, has been highlighted as posing selective pressure which may promote the spread of AMR in a large number of micro-organisms across various environments [23]. This ultimately results in the development of various life-threatening diseases in animals and humans, including in the bloodstream, urinary tract and respiratory tract inflammations [5,24].

The emergence of AMR can be classified into endogenous (such as gene mutation and recombination) or exogenous [such as horizontal gene transfer (HGT)] factors. The increase in endogenous AMR is primarily a result of selective pressure imposed by environmental changes, including severe environmental pollution, indiscriminate use of antimicrobials, etc., which influence the direction of microbial evolution by favouring specific adaptive trajec-

tories, leading to genetic mutation and/or recombination events that enhance survival in challenging conditions [25]. Many studies have pointed out that the indiscriminate use of antimicrobials can induce the emergence of exogenous AMR in the environment, humans and animals [6,7,21,26–29]. A review by Michalova et al. provided evidence that increased levels of tetracycline resistance in animal gastrointestinal tracts may be attributed to the long-term use of tetracyclines in veterinary medicine [30]. It has also been shown that livestock resistomes can be affected by factors such as diet [31], age [32,33] and living environment [34].

Organisms also acquire new AMR traits to adapt to environmental changes by obtaining ARGs from different microbial species, organisms or environments through HGT (Fig. 1). Successful HGT is driven by various assembled DNA elements consisting of potential mobile genes and mobile genetic elements (MGEs), such as plasmids, transposons, genomic islands, integrons, and integrative and conjugative elements (ICEs) [26,35], along with sufficient positive antimicrobial selective pressure and compatible recipients [36]. Forsberg et al., while looking at ARGs in soil and human pathogens, detected a large number of 100% identity ARGs in two isolates with large phylogenetic differences (non-pathogenic soil and human pathogens), demonstrating successful HGT without loss of functionality when genes were transferred between hosts in different environments [7,21]. This emphasizes the risks to One

Health posed by AMR, making it crucial to identify the origins of AMR, possible shared traits and ARG mobility between microbial species in human–animal–environmental interfaces.

1.2. The rumen and ruminant faecobiome as a reservoir of ARGs

Biologists have identified the rumen, the fermentation compartment of ruminant livestock, as a significant reservoir for ARGs and a potential source of novel AMR traits due to its diverse and complex microbial ecosystem [35,37,38]. This complex ecosystem may lead to competition, the production of new antimicrobial substances, and the significant development of resistance in this environment [21]. Auffret et al. [35] reported significant differences in the abundance of resistomes between concentrate-diet-fed and forage-fed cattle. Recently, it was found that the use of milk replacer in the diet of calves improved the abundance of *Prevotella* in the rumen, and changed the resistome positively compared with the use of normal milk [36]. Additionally, Gaeta et al. reported significant differences in microbiome diversity and increased resistome diversity in cows living on farms contaminated with heavy metals for long periods of time compared with cows living in normal environments [34]. In the context of a study on ruminant faecal samples, Smoglica et al. found that the gut microbiome composition of Italian ruminants varied across different ruminant species and land-use types, with ARGs showing a positive correlation with specific microbial taxa [42]. Additionally, Liu et al. [31] found that resistance in the faeces of cows was associated with their lactation diet, and that the diversity of the microbial community and resistance in their gut increased with age and dietary changes. These studies suggested that the microbial dynamics could be further influenced by nutrients [39] and diet, geographic location or spatial distribution [40], and environmental selective pressures [22], which jointly affect the evolutionary pathways of microbial communities. In addition, the close proximity of diverse microbial species facilitates HGT, enabling the dissemination of ARGs across different species [41].

A proportionately higher number of studies have focused on the microbial composition and resistance profiles of the faecal microbiome, which has been established as a reliable proxy for the rumen microbiome [44–48]. This is particularly significant as the faecal microbiome represents the most direct route for AMR transmission from ruminants to the environment, providing critical insights into the environmental dissemination of AMR through manure and run-off. For instance, Auffret et al. demonstrated that the bacterial communities in faeces closely mirror those in the rumen [35], thereby validating the use of faecal samples for studying rumen microbiota and AMR dynamics. Similarly, Shanks et al. [46] highlighted the efficacy of using faecal samples as a practical and non-invasive method for assessing microbial populations and ARGs present in the rumen for AMR surveillance in livestock. In comparison with faecal samples from ruminants, fewer studies have been conducted on AMR identification and characterization from the rumen [27,28,43] due to the complexities of studying this ecosystem. A limitation of using the faecobiome alone is that it may not fully capture the diversity of as-yet-unculturable rumen bacteria [49–51] that could play a critical role in the spread of AMR. A new concept is proposed here – the so-called ‘rumenofaecobiome’, defined as the combined microbiome of the rumen and the faeces in ruminant livestock, which encompasses the diverse microbial communities present in both the digestive tract (specifically the rumen) and the faecal matter of ruminants as a robust approach for holistically unravelling the microbiome and associated AMR dynamics in ruminants in a One Health context. This concept underscores the continuity, overlap and interaction between the microbial populations in the rumen and faeces of ruminants.

Bacterial genomes collated recently through the Hungate1000 project [52] and rumen metagenome assembled genomes [53,54] provided the foundation for studying the rumen microbiome and the occurrence and distribution of ARGs in rumen micro-organisms. Consequently, the colossal and diverse studies carried out on the rumen to date have made an important contribution to understanding this ecosystem. In addition, a metagenomic analysis of ARGs in bovine rumen conducted by Jing et al. [27] demonstrated differences in ARG diversity between different phyla. They observed that the prevalence of ARGs in different species of *Ruminococci*, one of the core rumen micro-organisms, was significantly uneven. This finding demonstrates that substantial variations in both the types and abundances of ARGs can exist between different microbial taxa, underscoring the remarkable complexity and extensive diversity of ARG distribution patterns within the rumen-associated microbial ecosystem.

Previous studies have shown that rumen bacteria not only carry ARGs but also carry transmissible elements with potential for HGT (e.g. *tetW* gene in *Blautia schinkii* and six other rumen isolates carried on a novel ICE) [26]. More recently, Faleiros et al. [55] found 31 associations (including virus-relevant and plasmid-relevant) between bacteria in the rumen of Nellore cattle and their identified MGEs with the Hi-C method that can capture the physical chromosome conformation. Moreover, Barbosa et al. [56] evidenced that genetic exchanges relevant to tetracycline ARGs occur between *Butyrivibrio fibrisolvens* strains isolated from the rumen. Hitch et al. [57] evidenced that ARG HGT between rumen micro-organisms and human pathogens is as flexible and frequent as the case of soil and clinical pathogens found by Forsberg et al. [7]. In this context, the rumen profiles can provide further direction and insights for studying HGT between species. Together, these studies demonstrate that the rumen harbours a diverse array of ARGs and MGEs, with significant variations in ARG distribution between species and phyla, highlighting the rumen and ruminant faecobiome as a critical reservoir for ARGs and a hotspot for HGT.

An analysis of 13 recent studies (Table 1) investigating the resistome of the rumen or faecal microbiomes across diverse countries and ruminant species over the past 15 years highlighted both commonalities and disparities in microbiome composition and AMR profiles. These studies, identified from NCBI PubMed using the keywords ‘rumen resistance/AMR metagenomics’ and ‘faeces/faecal ruminant resistance/AMR metagenomics’(2010–2025), were filtered to exclude non-metagenomic datasets, intervention-based studies, sequencing depths <10 million reads per sample, and inaccessible data. Only studies in which the data were generated and analysed as part of the same study were included.

Findings indicate that the core composition of the rumen and faecal microbiomes is highly consistent across ruminants, predominantly comprising Firmicutes, Bacteroidetes, Proteobacteria and Actinobacteria (Fig. 2A). This reinforces the strong functional and taxonomic similarities between the two microbiomes, supporting the need to study them collectively as the rumenofaecobiome to better understand AMR dynamics in ruminant livestock within a One Health framework.

The resistome profiles reveal that aminoglycosides, beta-lactams, multi-drug resistance, macrolide–lincosamide–streptogramin and tetracycline ARGs form the core resistome in the rumen and faecal microbiome. However, resistome diversity varied more between rumen studies than faecal studies (Fig. 2B). The faecal resistome remained more consistent across studies, likely due to reduced host–microbiome interactions post-excretion, and was primarily composed of the same dominant ARGs.

Additionally, rumen samples exhibited greater diversity of ARGs, including resistance genes for nucleosides, isoniazid, nitrofurans, acriflavine, tetracenomycin, nitroimidazoles and oxazolidinones (Fig. 2C). This highlights the unique complexity and diversity

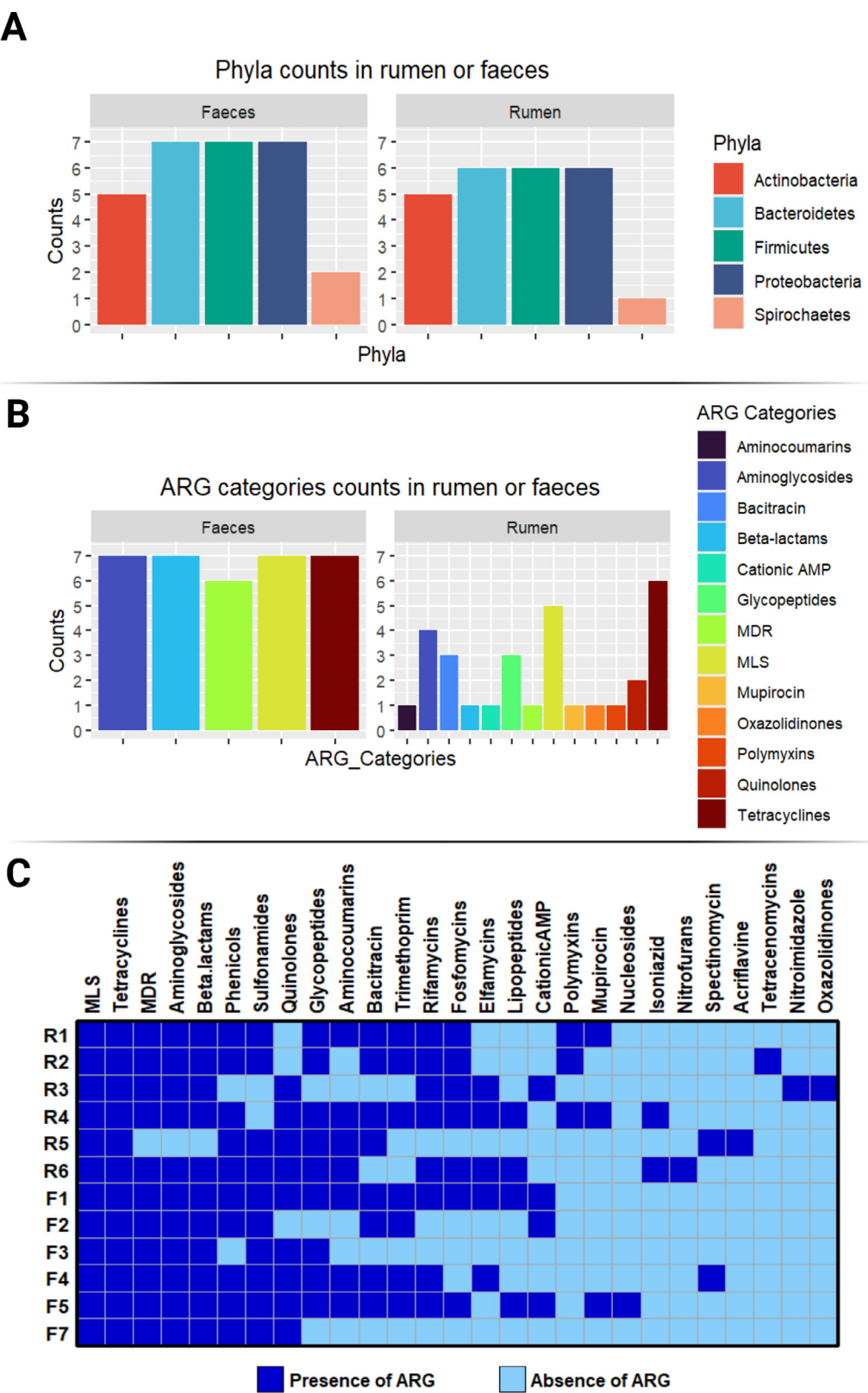


Fig. 2. Overview of the microbial community and resistome in ruminant faecal and rumen samples. (A) Predominant phyla across selected studies. (B) Predominant antimicrobial resistance gene (ARG) categories identified across studies. 'Count' on 'y' axes in A and B represents the number of studies reporting each phyla or ARG category in the top 4 and 5 observed, respectively. (C) Presence/absence of ARG categories identified in 12 of the selected studies (R, rumen; F, faeces). Study F6 was excluded due to incomplete resistome data. MDR, Multidrug Resistance; MLS, Macrolide–Lincosamide–Streptogramin Resistance.

Table 1
Summary of 13 metagenomic studies of rumen and fecal resistomes in ruminants (studies published between 2010 and 2025 that met the selection criteria).

Sample type		Animal species	Study location	Sequencing platform	Sequencing depth (reads per sample)	ARG database
Rumen	R1	Cattle [27]	UK	Illumina HiSeq 4000	82 million	ARGminer
	R2	Cattle [28]	Canada	Illumina HiSeq 2000	54.8 million	ARDB, CARD, NCBI-NR
	R3	Goat [33]	China	Illumina HiSeq 2500	48 million	MEGARes
	R4	Cow [29]	China	Ion Proton	30.8 million	CARD
	R5	Cattle [58]	France	Illumina HiSeq 2000	47.9 million	ARDB
	R6	Cow [59]	China	Illumina HiSeq 3000	61.1 million	CARD
Faeces	F1	Cow [31]	USA	Illumina HiSeq 4000	11.2 million	MEGARes
	F2	Cattle [60]	Canada	Illumina HiSeq 2000	117, 59 and 26 million	MEGARes
	F3	Cattle [61]	Canada	Illumina HiSeq 2000	54 million	MEGARes
	F4	Cattle [62]	USA	Illumina HiSeq 2000	46.3 million	Custom
	F5	Cow [63]	New Zealand	Illumina Novaseq S4	40 million	Abriicate, MEGARes
	F6	Cattle, bison, elk [64]	USA	Illumina NovaSeq 6000	14.59 million	MEGARes
	F7	Cattle, cow [65]	China	Illumina Hiseq X ten	87 million	CARD

ARG, antimicrobial resistance gene; ARDB, Antibiotic Resistance Genes Database; CARD, Comprehensive Antibiotic Resistance Database; NCBI, National Center for Biotechnology Information.

of the rumen environment, further emphasizing the importance of comprehensive resistome studies across both microbiomes.

1.3. The spatio-temporal dynamics of AMR evolution and transmission

Another phenomenon which is poorly studied in general, and particularly in ruminant livestock, is the spatio-temporal aspects of AMR [i.e. how ARGs move through microbial populations over time in different environments (e.g. from farms to waterways)]. ARGs, due to gene mutations and recombination triggered by various exogenous factors, can be transferred between generations during the asexual reproduction of prokaryotic micro-organisms, and remain present in livestock from infancy to adulthood. Also, ARGs can pass on to the next generation of livestock and humans through vertical transmission of their microbiome [66,67]. Although the resistome was not analysed, Furman et al. demonstrated that infant and adult ruminants shared the same core microbiome, with the composition of these microbes in infancy having a strong influence on the adult microbiota [68]. Huws et al. showed that, while the composition of the rumen microbiota changes over time, the core microbiome remained stable at a certain level to support the basic metabolic and degradative functions of the rumen [69]. The strong correlation between the rumen resistome and microbiome observed by Chai et al. [33] suggested that ARGs in the microbiome persisted throughout the ruminant's life, even if their prevalence decreased with age [70]. Patangia et al. reported that ARGs can be transferred from human mothers to fetuses through vertical transmission of the core microbiome [71]. It should be noted that vertical transmission described above [i.e. when AMR-related bacteria or genes are passed from either ruminant/human parent to their offspring (e.g. group B streptococci colonization via the placenta or exposure to the vaginal microbiome during birth [72])] is different from the vertical gene transfer (VGT). VGT refers to the transmission of genetic material from parent organisms to their offspring during reproduction. In the context of AMR, VGT involves, for example, bacteria passing ARGs carried on their chromosomes or plasmids to their progeny through cell division, leading to clonal expansion of resistant strains [73]. This process ensures that resistance traits are maintained stably across bacterial generations, contributing to the persistence of AMR in microbial populations.

At the same time, it is also crucial to study how ARGs move through microbial populations over time in different environments (i.e. patterns, processes or relationships that evolve over time within a particular space or geographic area simultaneously). Zhao et al. [40] modelled the geospatial prevalence of AMR for seven antibiotics in *Escherichia coli* and non-typhoidal *Salmonella* from

healthy food animals in selected developing countries, and found that tetracycline resistance was predominant. They predicted that, within the next decade, AMR to penicillins or tetracyclines will exceed the critical threshold (resistance rate >50%) in 78% of regions across South America and Africa, while AMR to penicillins or sulfonamides is projected to surpass this critical threshold in 77% of Asian regions. Mulchandani et al. [74] focused predominantly on food animals in developed European countries, with the highest predicted prevalence of AMR observed in *E. coli* strains across Romania and Italy, *Salmonella* across Hungary and Poland, and *Campylobacter* across Spain. Rosenkrantz et al. [75] conducted a study in northern Tanzania which demonstrated that the spatial relationships between farms and their management practices, such as antibiotic use, influence the distribution of antibiotic-resistant bacteria significantly. Furthermore, environmental stressors induced by climate change can alter the ecology of microbial communities significantly, fostering adaptive mutations and facilitating the HGT of ARGs [76]. This phenomenon presents a formidable challenge to the effective management and treatment of AMR. In the context of food safety, the rising global demand for animal protein has led to increased use of antimicrobials in livestock production, which, in turn, increases the risks of AMR infections [14]. Additionally, the transportation of livestock and their products across regions further facilitates the spread of AMR across borders, underscoring the critical need for enhanced biosecurity measures during animal movement [77]. He et al. [78] demonstrated that ARGs present in animal faeces can propagate through different environmental pathways and potentially transfer to human pathogens, indicating the interconnectedness of agricultural practices, environmental contamination and public health risks associated with AMR. These studies highlight significant regional differences and temporal shifts in the prevalence of AMR, underscoring the need for continuous monitoring to inform policy decisions. Therefore, exploring these spatio-temporal elements may improve understanding of AMR in ruminant livestock in the context of One Health.

2. Mechanisms of ARG function and transfer within and between bacterial cells

2.1. Potential AMR mechanisms in the rumenofaecobiome

Bacterial AMR mechanisms are generally classified into passive defence and active attack (Fig. 3). Passive defence includes basic physical barriers such as biofilms [79] (Fig. 3A); reducing intracellular drug accumulation by decreasing the permeability of the cell envelope (Fig. 3B) [80] and mediating antibiotic efflux (Fig. 3C);

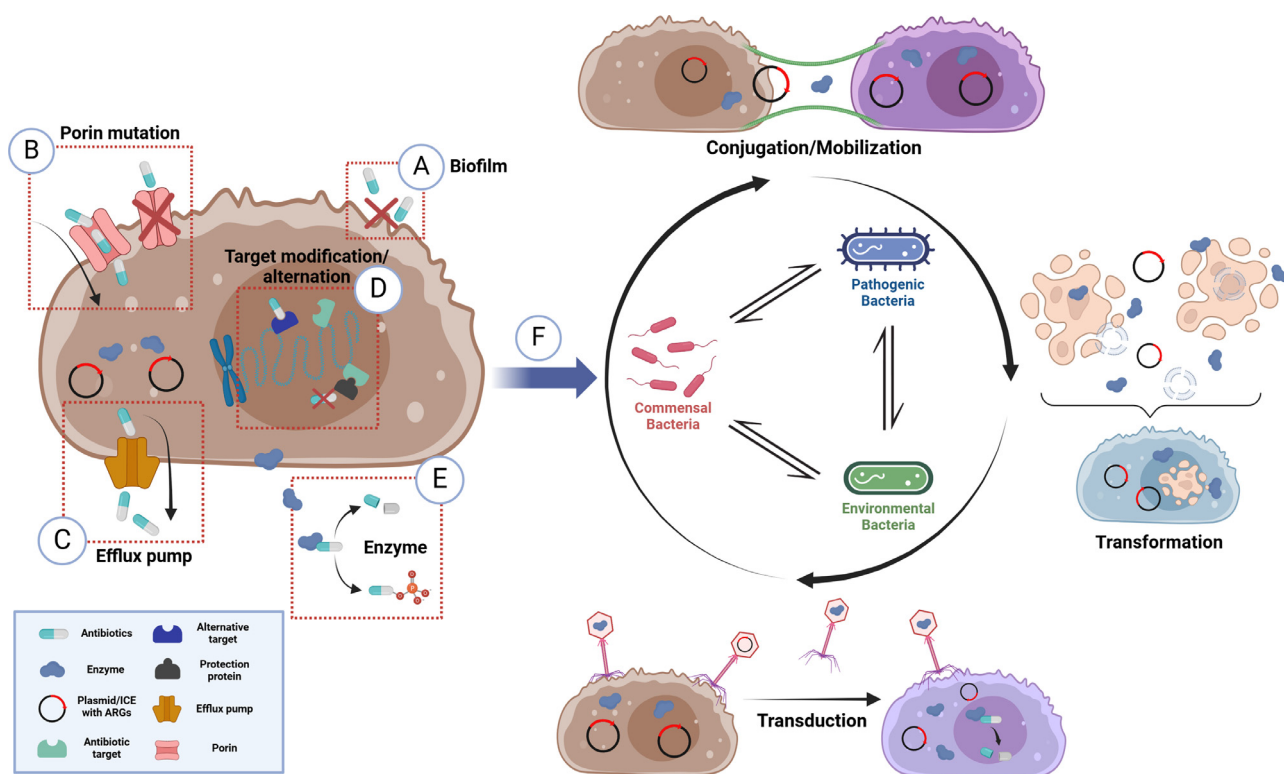


Fig. 3. Bacterial defence and antimicrobial resistance transfer mechanisms present in the rumenofaecobiome. (A) Physical barrier – biofilm [79]. (B) Alteration of membrane permeability (such as porin loss, membrane composition changes, etc.) [80]. (C) Antibiotic efflux pumps [92]. (D) Modification or replacement of the antibiotic-specific binding targets [81–85]. (E) Production of antibiotic-inhibiting enzymes [93,94]. (F) Intercellular transfer of antimicrobial resistance genes (ARGs) (conjugation [95,96], transformation [56,97] and transduction [98]). This is a simplistic illustration of the basic inter- and intramicrobial antimicrobial resistance mechanisms; however, the actual situations, which are more difficult to represent in a figure, could be more complex and multi-dimensional, with different mechanisms influencing each other. This figure was created using BioRender.

and modifying, altering or replacing cellular antibiotic-binding targets (Fig. 3D) [81–85]. Active attack involves modifying or disrupting the drug itself through enzyme production (Fig. 3E); and facilitating the spread of resistance via transformation, conjugation and transduction (Fig. 3F) [86,87].

The cell envelope in bacteria is a primary defence against antimicrobial drugs. Structural alterations can reduce permeability [80] leading to AMR, such as polymyxin resistance observed in a rumen bacteria study by Xue et al. [29]. Additionally, active expulsion of antimicrobials by energy-dependent transmembrane proteins such as efflux pumps [88,89] (Fig. 3C) contribute to AMR, especially in Gram-negative bacteria [90] predominant in the rumen [88,89]. Xue et al. found that 45% of ARGs related to elfamycin and streptogramin resistance in the rumen were associated with efflux pumps [29,43,91].

Antimicrobials that enter cells inhibit normal functions by binding to targets, causing growth termination or cell death. Bacteria counteract this by: using target-protective enzymes to spatially inhibit drug binding by target mutation [81] or dissociation [82] (Fig. 3D) [83]; employing target bypass mechanisms where alternative enzymes replace the original targets [84]; and protecting original targets with higher-affinity alternative targets or over-producing targets to dilute the effects of the drug [85]. Furthermore, bacteria can inactivate antibiotics by producing enzymes such as β -lactamase, which hydrolyses β -lactam antibiotics [93], and *tet(X)* tetracycline inactivase [33], which inactivates tetracycline drugs. *Tet(X)* genes, which have also been identified in the rumen [33], can spread horizontally via transposable elements, leading to multiple homologues globally [94] (Fig. 3E). Xue et al. (29) identified various ARGs in the rumen resistome linked to these mechanisms (Figs 2E and 3D). Predominant classes included ARGs for tetracyclines (92% in target protection), glycopeptides (95% in

molecular bypass), aminoglycosides (38% inactivation), multi-drug (53% target modification) and trimethoprim (59% target replacement). Eventually, ARGs and functional enzymes transfer between micro-organisms via transformation, conjugation or transduction (Fig. 3F), with orthologous ARGs evident across related species in the rumen environment. Extracellular DNA can also be taken up by transformation involving natural (e.g. in *Bacillus subtilis*, *Streptococcus pneumoniae*) [58] or induced/artificial competence (e.g. in *E. coli*) [97]. AMR dissemination can also occur by conjugation/mobilization via plasmids [95], ICEs [96] and transduction mediated by bacteriophages [98].

2.2. MGEs are key drivers of ARG spread in the One Health framework

MGEs play an important role in the dissemination of ARGs in the One Health framework as they are involved in the intracellular and intercellular transmission of ARGs. These include small elements, such as insertion sequences (IS) and transposons (Tn), and large elements, such as plasmids and even bacteria themselves (Fig. 4). Unfortunately, AMR-relevant MGEs in the rumen are understudied.

2.2.1. Insertion sequences and transposons

IS are the smallest MGEs, typically consisting of one or two transposase proteins (Tnp) surrounded by terminal inverted repeats (IR). Although not considered to carry ARGs in their sequences, IS can form circular Tn (Fig. 4A) during transposition, allowing movement within the cell or insertion into other DNA molecules mediated by Tnp. Composite transposons, containing ARGs flanked by identical IS, transfer between DNA molecules (Fig. 4B). The IS6 family, particularly IS26 (Gram-negative), IS257

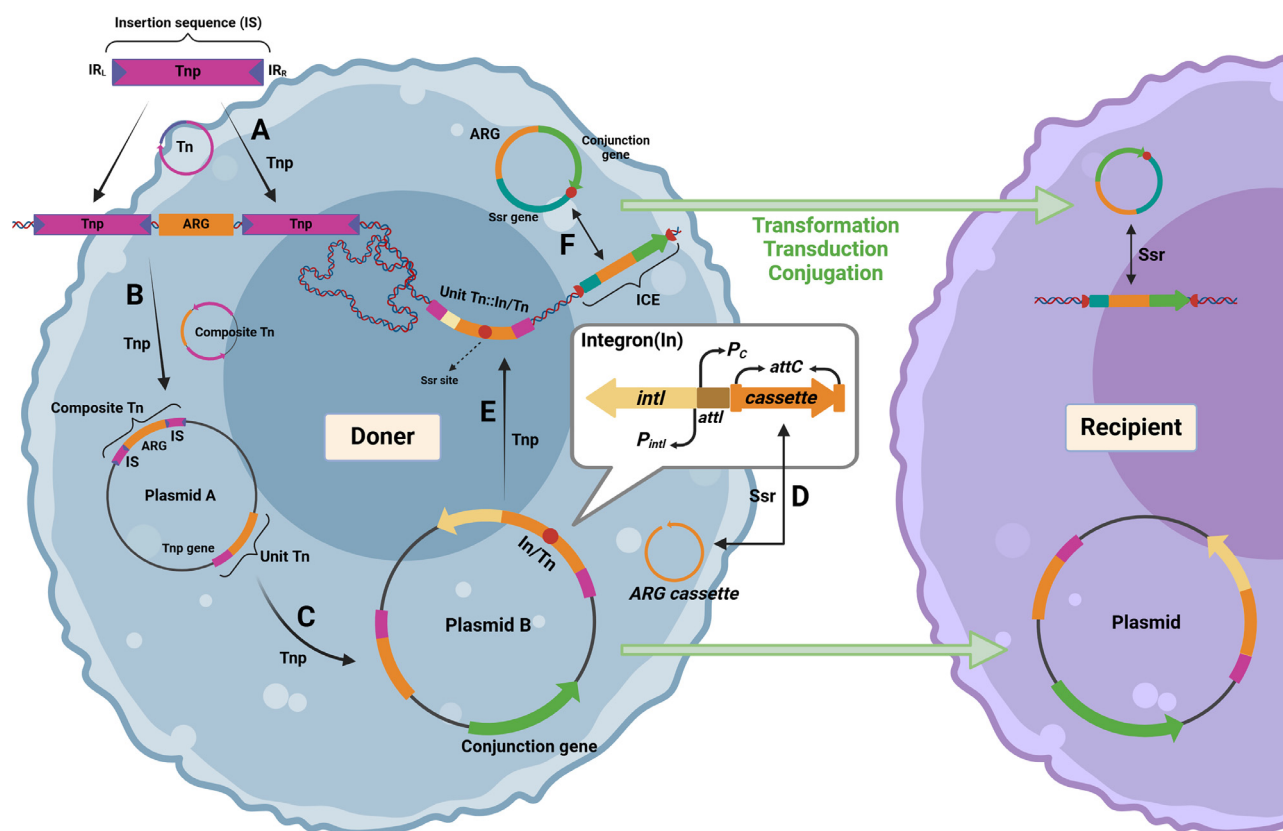


Fig. 4. Intra- and intercellular mechanisms of mobile genetic elements (MGEs) in bacteria. The bacterium in blue represents the donor cell, while the bacterium in purple represents the recipient cell. MGE mobility is mainly mediated by transposase protein (Tnp) or site-specific recombinase protein (Ssr). The intracellular transfer processes identified by black arrows are mainly concluded as six types. (A) Free insertion sequences (IS) form as a circular transposon (Tn). (B) Insertion of the same IS on both sides of the antimicrobial resistance gene (ARG) allows them to excise and combine as a composite Tn [99]. (C) A single IS carries an ARG which forms as a unit Tn [100]. (D) The ARG cassette on integron can only detach and then intracellularly bind to another integron (In) if without the mediation of Tn [101,102]. (E) Tn carries an ARG that is inserted on an In and transfers as an In/Tn complex [103]. (F) Integrative and conjugative elements (ICEs) are a type of MGE integrated into chromosomes, and can transfer to the recipient as a loop element during cell conjugation [96]. Finally, plasmids and ICEs carrying ARGs can achieve intercellular horizontal gene transfer via bacterial transformation, transduction and/or conjugation processes, which are annotated by the light green arrows [104]. This figure was created using BioRender.

and IS1216 (Gram-positive) [99], impacts the transmission of ARGs significantly [62]. Unit transposons, flanked by IR, instead of IS pairs, carry ARGs independently (Fig. 4C), and generally include a *tnpA* transposase gene, a *tnpR* resolvase gene and one resolution (*res*) site, facilitating repeated travel between chromosomes and plasmids [100]. HGT mediated by Tn or plasmids has been demonstrated in lactic acid bacteria using in-vitro filter mating and in-vivo mating techniques [105]. Likewise, putative conjunctive Tn was found to be engaged in HGT processes by inverse polymerase chain reaction (PCR) in the genome of *Prevotella bryantii* B14 isolated from rumen [106].

2.2.2. Integrons and gene cassettes

Gene cassettes, small MGEs carrying ARGs, typically consist of an open reading frame (ORF) with *attC* recombination sites on both sides [101,102]. The *attC* site on the gene cassette can recombine with the *attI* recombination site on the integron catalysed by an integron integrase, allowing the gene cassette to be inserted (Fig. 4D). Non-replicative and fragile, gene cassettes integrate into integrons via integron integrase, allowing ARG propagation (Fig. 4E). Integrons, classified into *IntI1*, *IntI2* and *IntI3* groups, vary by homologous *attI* sites. Class I integrons, the most common in AMR studies, carry cassettes with aminoglycoside and trimethoprim resistance genes, found in *Enterobacter* spp. [103], *Enterococcus faecalis* [107] and *E. coli* [108]. Class II integrons, carrying ARGs, have been detected in *Salmonella* spp. [109] and *E. coli* [108], while the less common Class III integrons carry ARGs in few species, in-

cluding *E. coli* [108]. These species and their integrons have also been found in the rumen collection [52], although their functions in the rumen are underexplored.

2.2.3. Resistance plasmids

Plasmids replicate independently of the chromosome, inheriting AMR traits through reproduction and facilitating vertical and horizontal transfer of ARGs. Plasmid backbone genes ensure heritability and mobility, while accessory genes confer specific properties, contributing to phenotypic diversity [99]. Plasmids with similar backbones may carry different accessory genes, and homologous ARGs can insert into various plasmid backbones, diversifying the AMR environment [104]. For example, the pRR14 plasmid in *Bacteroides rumenicola* carries tetracycline resistance genes [110]. Faleiros et al. [55] used Hi-C to integrate MGEs with rumen bacterial genomes from Nellore cattle, identifying 19 plasmids integrated into host genomes and 33 non-integrated plasmids. These integrated plasmids contained ARGs with >90% identity tetracycline (*tetQ*, *tetW*) and β -lactam (*blaACI-1*) resistance genes, with several of the non-integrated plasmids also associated with host genomes [55].

2.2.4. Gene islands

ICEs are removable gene islands in bacteria [99] that can typically integrate into the host chromosome and replicate, segregate and undergo cytokinesis as part of the chromosome (Fig. 4F) [96,99]. The mechanism of conjugation of ICEs between cells is described in Fig. 5. Sabino et al. identified a putative ICE carrying the *tetW* tetracycline resistance gene from *Blautia schinkii* DSM

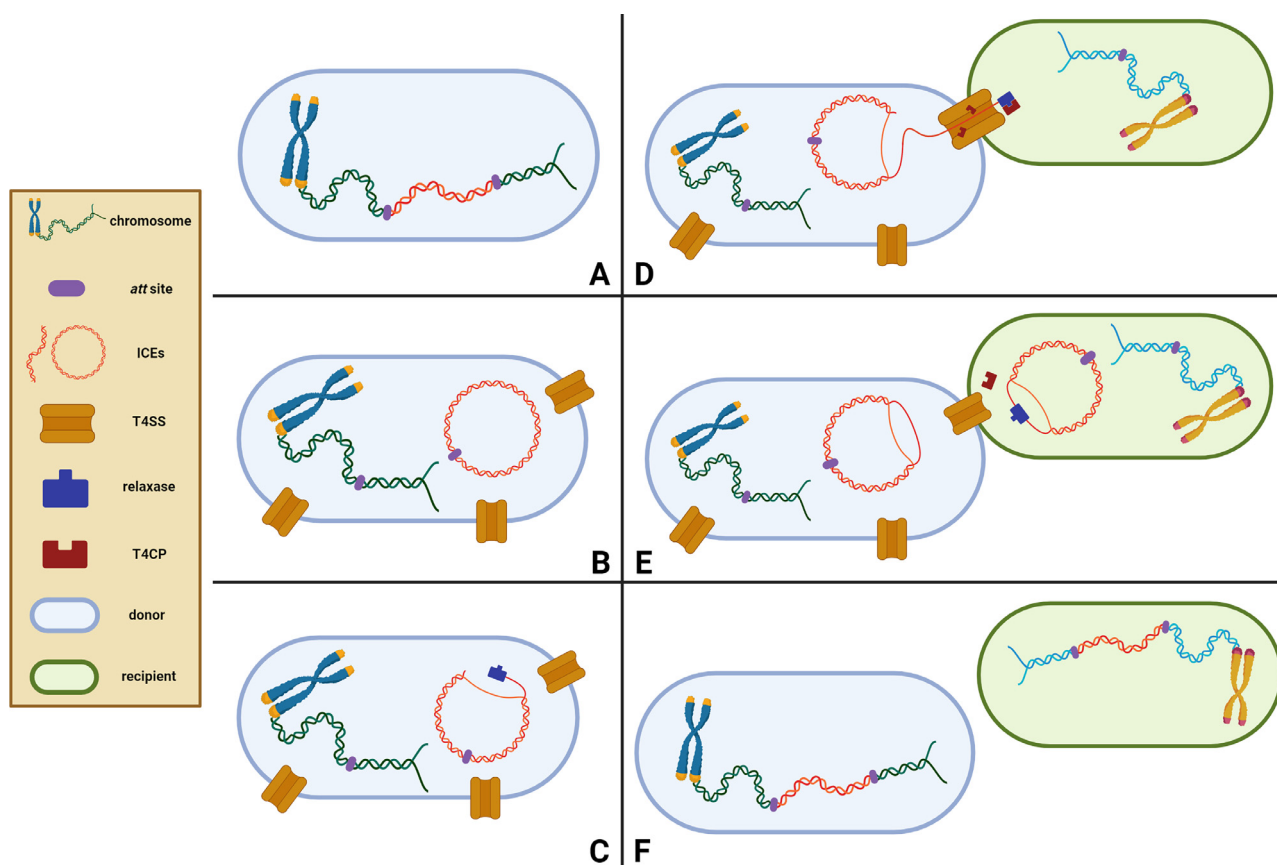


Fig. 5. Mechanism of integrative and conjugative elements (ICEs) in bacteria. The representative components in the illustration are identified in the yellow box on the left. (A) ICE inserts at the defined attachment site (*att*) on the chromosome [115]. However, the majority of the ICE genes are only expressed with activation or induction. (B) The induced ICE excises from the chromosome and forms a circular double-strand DNA (dsDNA). Some of the associated ICE-encoded proteins assemble into cylindrical mating pores named 'type IV secretion systems' (T4SS) which can span the envelope of donor and recipient [116]. (C) The ICE-encoded relaxase cuts and joins a strand of the dsDNA by recognizing the homologous origin of transfer (*oriT*) on ICE [117], forming a single-strand-DNA-protein complex called 'transfer DNA' (T-DNA). (D) T-DNA couples to the type IV coupling protein (T4CP) and is then pumped into the recipient by T4SS [116]. (E) T-DNA re-forms as a conjugative dsDNA in the recipient. (F) The conjugative dsDNA inserts into the chromosome in the recipient cell, and the remaining original dsDNA template in the donor regenerates dsDNA by rolling-circle replication, and then integrates back into the original chromosome [96]. This figure was created using BioRender.

10518 isolated from the rumen [26]. This newly discovered ICE genetic organization was found to contain putative genes encoding for conjugation proteins, secretion proteins, cellular cycle controlling proteins, recombinases and other basic accessory proteins associated with MGEs. In addition, some ICEs have been discovered that can replicate autonomously [111], making the boundaries between ICEs and plasmids ambiguous. The broad definition of genomic islands also includes integrative mobilizable elements, which require helper functions for conjugation, and elements transferred horizontally via phage-mediated transduction [112]. In 1988, a miniature inverted-repeat transposable element (MITE) resembling IS elements was discovered in *Neisseria gonorrhoeae* without transposase genes for autonomous transposition [113]. MITEs were also found flanking integron fragments carrying cassette arrays in *Acinetobacter johnsonii* [114]. The presence of MITEs and Tn3-derived inverted-repeat miniature elements (TIMEs) may explain instances where complete IS or Tn elements are absent for ARG transposition, although they are challenging to identify and remain understudied in the rumen environment.

3. Unravelling AMR in the rumenofaecobiome

3.1. Overview of the current status

In general, the rumen bacteria known to date are mostly from well-explored culturable groups or the genomic reconstruction

of unculturable taxa. Although advances in genomic technologies have advanced our understanding of microbial systems, isolating and culturing as-yet-unculturable rumen bacteria remain essential for unravelling new insights into their ecological role in AMR. Uncultured anaerobic rumen bacteria may include those for which little effort has been made to isolate them in culture, or fastidious organisms unable to grow under less complex culture conditions but which still have the potential to be cultured [118]. This generally means that these as-yet-unculturable bacteria are poorly studied, resulting in the belief that they are 'unculturable' [119]. For example, Hess et al. found that only 0.03% of rumen metagenomic scaffolds matched (>90%) available genomes in GenBank [120]. Similarly, Creevey et al. noted that many core cultured rumen bacterial species lacked reference genomes and, although not novel, were still classified as unculturable [121]. In 2017, Huang et al. analysed the bacterial sequences from sheep rumen samples in two different regions of China, and compared them with the GenBank database. For half of the cultures, no matches in the databases were found, especially for sheep living in harsh plateau environments [122]. Similarly, Wu et al. reported a significant taxonomic gap between the rumen metagenomes and the genomes from the pure ruminal isolates, which indicated that a large proportion of these microorganisms remain uncultured and therefore understudied [123].

This gap in reference genomes and incomplete databases hampers our understanding of rumen microbial interactions and the resistome [124]. Christian et al. [125] found that aminoglycoside

and tetracycline ARGs cloned from uncultured bacteria were inconsistent with the sequences of all homologous ARGs in the reference database, suggesting that the unculturable bacterial population harbours a large number of potentially novel ARGs which may further improve our understanding of ARG carriage and transmission from ruminants in the One Health context. These studies underscore the importance of how studying uncultured populations could provide unique insights into the risks of AMR transmission compared with existing studies that focus primarily on culturable faecobiomes alone. Although the AMR profiles of as-yet-uncultured bacteria of the rumenofaecobiome remain largely unexplored, new insights into their potential for biotechnological exploration provide an insight into what we might expect to learn about the resistome in ruminants by using a more comprehensive approach. For example, novel antibiotics [126,127] have been discovered in as-yet-unculturable soil microbes linked to the rumenofaecobiome, emphasizing their relevance in the context of One Health. Ling et al. [127] successfully cultured and isolated previously unculturable microbes by simulating a natural environment using an isolation chip (iChip) composed of microdiffusion chambers [128,129], supplemented with iron-chelating siderophore growth factors, leading to the discovery of teixobactin [130]. Therefore, filling in these missing knowledge gaps through intensive rumenofaecobiome research is crucial work to increase the repertoire of as-yet-unculturable rumen bacteria, and improve our understanding of their role in AMR carriage and transmission.

Recent advances have included the 'Hungate1000' project with 410 rumen microbial genomes, 96.4% of which were not recorded in the Ribosomal Database Project. More than 70% of these rumen bacteria species did not have representative live culture stocks in international culture collections [52]. Stewart et al. further enhanced public databases of rumen microbial genomes, providing vital resources for biomass-degrading enzyme discovery [54]. These efforts laid the groundwork for studying ARGs in rumen micro-organisms. Current global initiatives, such as the Rumen Gateway project [131], aim to cultivate as-yet-unculturable rumen microbes using, among other methodologies, traditional culturing methods (direct plating, dilution to extinction, enrichment culture) and advanced high-throughput methodologies (microencapsulation [132] and fluorescence-activated cell sorting [133]). These would fill crucial gaps in our understanding to supplement current rumen microbial culture gaps, thus providing a valuable resource for a better understanding of this ecosystem.

3.2. Current methods for unravelling AMR in the rumenofaecobiome

3.2.1. In-silico omics-based methods for ARG detection

3.2.1.1. Overview of genotypic omics-based technologies. In order to capture the ARGs in uncultured populations effectively and analyse the AMR profiles in the rumen microbiome comprehensively, it is crucial to investigate the origins and locations of ARGs, understand how they are expressed and transferred across organisms and environments, and examine their phenotypic behaviour *in vivo*. Next-generation sequencing technologies based on genomes, transcriptomes, proteomes and metabolomes are pivotal to achieving this. Among these, whole-genome sequencing (WGS), a common culture-dependent approach, and metagenomic sequencing (MGS), a culture-independent method, are the most commonly used sequencing techniques for detecting ARGs (Fig. 6A) [134].

Genomics and bioinformatics analysis allow for the exploration of genetic material in cultured micro-organisms. By aligning the sequencing reads or assemblies against a reference database, researchers can annotate the location of the target gene precisely, and determine its similarity to known reference sequences. Although they have high operational complexity and limited accessibility, advanced high-throughput methodologies such as microen-

capsulation [132] and fluorescence-activated cell sorting [133] automate culture-dependent approaches for pure culture genomics study, and have potential for unravelling genomic information from single cells.

On the other hand, metagenomics provides a comprehensive overview of the majority of the genetic components within entire microbial communities, expanding gene databases significantly. Nevertheless, metagenomics has limitations, including the partial loss of microbial details, reduced suitability for in-depth analysis of small-scale samples, and poor experimental reproducibility due to the complex and stochastic nature of sample compositions [134]. Additionally, metagenomics enables the extraction of valuable genomic information from the uncultured or as-yet-unculturable microbial populations, thereby facilitating the design of targeted cultivation strategies tailored to their specific physiological requirements [135]. Both WGS and MGS have distinct advantages and drawbacks, necessitating a careful choice of method based on the specific research objectives. In many cases, combining the two approaches may provide a more comprehensive understanding of AMR in the rumen microbiome.

Genomic and metagenomic studies often rely on assembly-based or read-based bioinformatic approaches. Read-based tools such as Bowtie2 [136] or Burrow-Wheeler Aligner (BWA) [137] directly align short DNA sequences to reference databases, enabling faster and simpler ARG detection. However, this method depends on database completeness, risking false positives and missing genetic content from unculturable bacteria. In contrast, assembly-based methods (Fig. 6A), such as SPAdes [138], use De Bruijn graph-based assemblers to reconstruct genome sequences [87], predict protein-coding regions, and annotate AMR determinants by comparing them with ARG databases via tools such as the basic local alignment search tool (BLAST) [139] or DIAMOND [140]. While useful for identifying novel ARGs and MGEs, this approach is time-consuming, computationally expensive, and prone to gene content duplication or loss [141]. A key consideration in ARG annotation, especially when using assembled data, is the accurate prediction of coding sequences (CDSs). While ORFs are defined by start and stop codons [138], they do not all encode functional proteins [142], making accurate ARG identification complex. Variability in start codon usage [134] and stop codon recognition [143] can introduce biases, as seen in *Mycoplasma*, where the TGA stop codon was recoded as tryptophan, leading to truncated ORF predictions while true CDSs extended correctly [144,145]. These factors can affect the accuracy of ARG annotation, underscoring the importance of robust bioinformatic approaches; for example, Ma et al. [28] applied both approaches to analyse AMR profiles in metagenomes and metatranscriptomes from the same rumen samples. Their findings revealed that, except for tetracenomycin, the read-based method identified a greater diversity and quantity of ARGs compared with the assembly-based approach.

Overall, assembly-based methods are well suited for analysing novel genomes and resolving structural variations, whereas read-based approaches excel in rapid screening of known genes and detecting small-scale genetic variants [134]. However, both approaches must account for the inherent uncertainties in gene annotation to improve the accuracy of detection of AMR.

Transcriptomic analysis usually performed after RNA sequencing allows for the characterization of ARG expression within a specific context, offering insights into their behaviour under defined conditions. It offers higher sensitivity and a broader dynamic range compared with traditional methods such as microarrays which rely on specified probe sequences. It links genomic data with expressed proteins, providing insights into ARG expression and regulation [159]. Single-cell RNA sequencing allows for cellular-level transcriptome comparisons, essential for culture-dependent AMR analysis. Spatial transcriptomics, introduced in 2016, can visualize

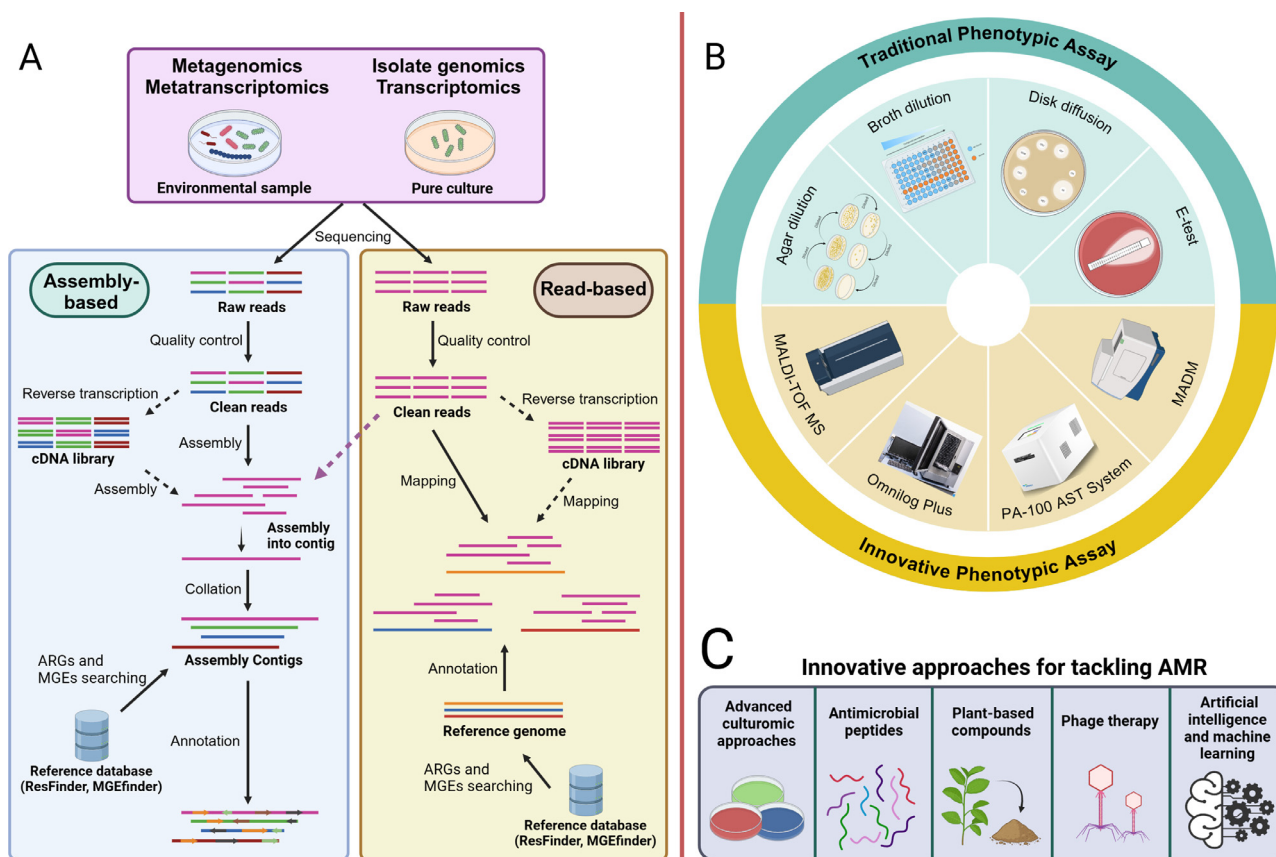


Fig. 6. Methods and current advances for unravelling the resistome. (A) In-silico assembly-based and read-based genomic approaches in metagenomics/metatranscriptomics and genomics/transcriptomics studies. The black dashed arrows represent the specific steps (reverse transcription of RNA and complementary DNA library establishment) for both metatranscriptomics and transcriptomics studies. The purple dashed arrow illustrates the clean reads from the isolate genomics studies, which can also be performed by assembly-based approaches [134]. (B) The current traditional and innovative advances for phenotypic assays for antimicrobial resistance (AMR) study [146–153]. (C) Innovative approaches for the diagnosis and amelioration of AMR One Health risk in livestock rumenofaecobiomes [154–158]. The figure was created using BioRender. ARG, antimicrobial resistance gene; MGE, mobile genetic element; MALDI-TOF MS, matrix-assisted laser desorption/ionization – time-of-flight mass spectrometry; AST, antimicrobial susceptibility test; MADM, multiplexed automated digital microscopy.

spatial distribution and mRNA abundance from the RNA sequencing data of mouse brain and human breast cancer [160], with sci-Space extending this to single cells in 2021, although mainly in mammalian embryonic studies [161]. Spatial transcriptomics can also identify expression differences during biofilm formation, elucidate key genes and pathways involved in this process, and compare transcriptional variations among different species during biofilm development [162–164].

Metatranscriptomics like metagenomics helps to profile the ARG expression level of entire ecosystems by matching mRNAs with target ARGs using alignment tools such as Bowtie2 [54], BWA and BLAST, and annotations from Gene Ontology, Kyoto Encyclopedia of Genes and Genomes (KEGG), COG and Swiss-Prot [134]. Metatranscriptomics can detect the actual expression levels of ARGs, thereby providing a more accurate reflection of AMR phenotypes compared with metagenomic approaches. Consequently, changes in the transcriptome provide a more direct reflection of how antibiotic treatments impact resistant microbial communities, highlighting its significant applications in clinical and veterinary medicine. For example, in the study by Korry et al., while antibiotic treatment impacted AMR in the mouse gut microbiome at both the DNA and RNA levels, changes in the RNA transcriptome reflected the administered drug more specifically, demonstrating greater drug specificity [165]. Transcriptomic approaches also helped to identify the *tet(W)* gene as the most expressed tetracycline resistance gene in ruminants [42,87,88], highlighting its prevalence in the ruminant AMR landscape.

3.2.1.2. Tools and reference databases commonly used for ARG studies. Once sequencing reads are obtained and/or assembly of sequencing reads has been completed by the methods outlined above, ARG profiling is subsequently completed through a range of ARG reference databases. Reference databases frequently utilized in ARG studies provide comprehensive repositories of known ARGs, enabling researchers to identify and analyse ARGs across diverse microbial populations, although these databases can vary in terms of coverage, accuracy, and the inclusion of novel or less-characterized ARGs. ResFinder [166,167], including the updated ResFinderFG [168], is a web-based tool that compares sequence data against a curated database of known ARGs. It utilizes both genomics and metagenomics approaches to enhance the identification of ARGs, allowing for more comprehensive detection without relying solely on reference databases. It matches raw reads directly to its database and predicts AMR phenotypes from genotypes using functional ARG metagenomic datasets. However, ResFinder and ResFinderFG only accept nucleic acid sequences as input, whereas Resistance Gene Identifier (RGI) uses the Comprehensive Antibiotic Resistance Database (CARD) [169], and accepts both DNA and protein sequences as input. Additionally, RGI integrates Antibiotic Resistance Ontology with acquired and mutated ARG data from both genomic and metagenomic studies, and can predict both intrinsic, targeted and mutated ARGs by extrapolation from a protein homologue model (specific for functional homologues of AMR proteins) and a protein variant model (specific for AMR-associated mutations).

Other databases, such as the Antibiotic Resistance Genes Database [170], Lactamase Engineering Database [171], ABRicate [172], Antibiotic Resistance Gene-ANNOTation (ARG-ANNOT) [173] and National Center for Biotechnology Information (NCBI) database [174], are also often used. Specialized tools such as AMRFinder-Plus [175], which uses an NCBI protein-based database, combine BLAST with Hidden Markov Models (HMMs) to identify known and novel ARG sequences, achieving a 98.4% match with phenotypic susceptibility tests and ResFinder results. AMR++ v3.0 [176], a high-throughput pipeline for large-scale metagenomic data, uses the MEGARes database to detect diverse ARGs and confirm variants with new algorithms. The PanRes database [177], published in 2023, integrates multiple ARG databases including ResFinder, ResFinderFG, CARD, MegaRes, AMRFinderPlus, ARG-ANNOT [173], the 'CsabaPal' collection [178] and BacMet [179], enhancing detection across various research needs and offering a comprehensive toolset for ARG analysis. Although some of these ARG databases are not actively maintained, or are archived, obsolete or outdated (e.g. ARG-ANNOT and ARGminer [180]), the existing bioinformatic tools and databases remain crucial in advancing our understanding of ARGs and AMR dynamics across diverse microbiome contexts, and there is a shift towards more comprehensive and up-to-date resources in the field.

3.2.2. In-silico omics-based methods for assessing ARG transmission

Beyond detecting ARGs, which are ubiquitous in many environments, identifying and annotating MGEs are crucial for understanding the transmissibility of ARGs. Tools such as MGEfinder [181] and MobileElementFinder [182] detect MGEs in bacteria, using databases with genetic information on known MGEs, including IS, plasmids, ICEs, etc. MobileOG-db [183] organizes protein families from eight MGE databases into functional groups such as integration/excision and replication, reducing false positives from database redundancies. Nonetheless, the use of specific databases is necessary to accurately identify and confirm various MGE types due to their distinct mechanisms and genetic content. ISFinder [184] focuses on bacterial IS elements, using BLAST analyses to categorize candidates and provide comprehensive information on sequences and distribution. IntegronFinder [185] detects integrons with high sensitivity using HMM profiles [112] for integron-integrase and covariance models [186] to identify attC sites, and can also annotate known P_{int} and P_c promoters, attI sites and their ARGs. PlasmidFinder [187,188] identifies plasmid sequences in Enterobacteriaceae and Gram-positive bacteria, with data from thousands of plasmids in the NCBI database. ICEberg [189] archives detailed information on ICEs, enhanced by its latest version in 2024 [190]. PHASTEST [191], a phage identification tool, offers gene annotation and visualization, improving on PHASTER [192].

3.2.3. In-vitro AMR phenotyping methods

Complementary to in-silico multi-omics, in-vitro methods such as agar and broth dilution tests confirm AMR phenotypes by measuring the minimal inhibitory concentrations (MICs) of antimicrobial agents (Fig. 6B). Microbial cells are inoculated on agar or in 96-well plates with diluted antibiotics, with results assessed visually or by automated systems after incubation [146]. Although accurate, these methods are mostly culture dependent, labour intensive and have short shelf lives. Disc diffusion, a simpler and cheaper method, involves spreading bacterial suspensions on Mueller–Hinton agar with antibiotic discs, measuring inhibition zones against the standard published by the Clinical and Laboratory Standards Institute [147]. However, it provides only categorical results [148], and remains underexplored for anaerobes [193,194] or for comparing different antibiotics due to varied diffusion rates.

E-tests combine the simplicity of disc diffusion with the quantitative benefits of dilution methods, using strips with gradient antibiotic concentrations to determine MICs, but are much costlier [195].

Automated or semi-automated rapid antimicrobial susceptibility test (AST) techniques address the time-consuming nature of traditional ASTs (Fig. 6B). Multiplexed Automated Digital Microscopy performs species identification and AST of clinical samples in 6 h [149]. The PA-100 AST System from Sysmex Astrego, which won the Longitude Prize on AMR in June 2024, can identify pathogens in 15 min and complete AST in 45 min [150]. The Omnilog Plus ID (Biolog, USA) system accommodates up to 50 dedicated 96-well plates for large-scale phenotyping. It measures bacterial respiration by changing the colour of tetrazolium redox dye, capturing digital images, and plotting analysis curves [153]. This system has been validated for drug resistance phenotypes in *Staphylococcus aureus* [151] and is being optimized for anaerobes. Moreover, matrix-assisted laser desorption time-of-flight mass spectrometry has also shown its innovative potential to diagnose AMR in clinical studies [152]. However promising AST technology is not widely available for rumen microbes, and its reliability remains to be seen.

4. Future research directions for combating AMR in the rumenofaecobiome

4.1. Understanding the challenges of AMR

4.1.1. Tracing the origins and evolution of ARGs: perspectives and challenges

Deciphering the origins and evolution of the ARGs is crucial for understanding their mobilization, the mutation mechanisms that drive their evolution, and the mechanisms by which AMR spread across microbial populations, especially those in complex niches such as the rumenofaecobiome. Surprisingly, by comparative analysis of the genomes of sufficiently large numbers of bacterial studies, Ebmeyer et al. [37] found, in 2011, that most pathogenic microbes identified with ARGs were not their producers but rather their recipients and transmitters. This highlights that these pathogens were not the original hosts for the ARGs discovered, but were mostly from environmental intermediates. On the other hand, it has been shown that although most micro-organisms intrinsically harbour significant amounts of ARGs, only a few of them expressed resistance to antimicrobials and are able to transfer these ARGs to different species through HGT (acquired resistance), with many of the ARGs being non-expressed or involved in other functions (such as metabolism, biosynthesis, ribosome recycling, etc.) in the original hosts [196,197]. Moreover, the phenomenon of cross-resistance, wherein micro-organisms develop resistance to multiple antimicrobial agents sharing similar mechanisms of action, can lead to discrepancies between genotypic and phenotypic resistance profiles [198]. This not only complicates the interpretation of resistance data, but also poses significant challenges in designing effective therapeutic strategies.

4.1.2. Addressing methodological biases in AMR studies

Non-biological biases arising from methodological differences must be considered carefully in AMR studies. These biases can stem from variations in the assembly tools or reference databases used [199]. Differences in methodologies, as well as the composition and sources of databases, can lead to distinct focal points, when employing different tools for the same analysis can generate differing results. As sequencing technology becomes increasingly accessible, the sources contributing to public databases have grown more diverse. Different research centres often bring varying levels of expertise and biases to sequencing practices, and the specific protocols or even version changes in sequencing tools they employ can introduce unforeseen variations in annotation outcomes. This

heterogeneity ultimately leads to the incorporation of mixed biases into genomic databases [200].

Even within the same method, discrepancies can arise due to variations in parameters or operational practices. For instance, sequencing different microbial species presents varying levels of difficulty, and during the metagenomic sequencing process, specific taxa may be emphasized preferentially during DNA extraction, amplification and assembly, resulting in systematic deviations from true values [201]. Additionally, biases may be introduced by differences in the sample sources, models or systems of equipment used [202]. These methodological disparities were evident in studies such as Hitch et al. [57], where daptomycin ARG was identified as the most common ARG type in the rumen, while tetracycline ARGs were reported as the most prevalent by Sabino et al. [26]. Similarly, *tet(W)*, which codes for tetracycline resistance, has been widely recognized as the most prevalent and expressed ARG in the bovine rumen [26,28], whereas *TcmA*, which codes for resistance against tetracenomycin C, was identified as the core ARG in the buffalo rumen in work by Sun et al. [32].

Additionally, most current microbial AST methods are difficult to perform within anaerobic chambers. For example, measuring and adjusting the concentration of bacterial suspensions for microbroth dilution and E-test assays typically require alternating between aerobic and anaerobic conditions. This not only increases time and labour costs, but also increases the risk of anaerobic cell death due to exposure to oxygen. In fact, in comparison with aerobic counterparts in clinical pathogens, there are currently no standardized methods for testing phenotypic resistance in strict and especially slow going and fastidious anaerobes such as those from the rumenofaebioime. The European Committee on Antimicrobial Susceptibility Testing, for instance, only has breakpoints for a handful of anaerobic bacteria, which are mainly of human origin [194].

4.1.3. Culturing limitations for rumenofaecobiome research

Many studies are employing culturomics to discover new microbial taxa and to detect new ARGs, particularly in the livestock industry. For example, *NpmC* methyltransferase, which was discovered in unculturable bacteria within human and animal gut microbiomes in China and Canada, was found to be capable of neutralizing aminoglycosides [203]. In addition, Liu et al. discovered that the plasmid-mediated *mcr-1* gene in pigs could confer polymyxin resistance in bacteria, and be transferred horizontally via plasmids [204]. Findings from these culturomic studies provide new avenues for enhancing public safety in terms of both human and animal health, as well as informing strategies for environmental management in the context of AMR.

In the context of the rumen microbiome, our understanding of its roles in AMR carriage and transmission is limited as many rumen bacteria are not yet available in culture. Increasing the culturability of as-yet-unculturable rumen bacteria through new culturomics approaches, including design of new media and the dilution to extinction method [205–207], is crucial for understanding and addressing AMR in this ecosystem, as these bacteria may serve as uncharted reservoirs for understanding and ameliorating AMR transmission (Fig. 6C). Nonetheless, there are still significant challenges in culturing many micro-organisms from the rumenofaecobiome. Firstly, certain microbes have strict requirements for growth, including atmospheric composition, temperature, pH, salinity and other factors, as well as nutritional media substrates. With current techniques, it is difficult to accurately determine all the specific ingredients and growth conditions needed for culturing a bacterium [208], as isolating a specific species or genus by designing new media and simulating its natural growth environment in the laboratory remains a highly challenging and complex task. Hobson M2 media [209] incorporates clarified rumen fluid

to recover ruminal nutrients, while Kenter's media has demonstrated greater potential for replicating natural rumen conditions [205,206]. Nevertheless, many micro-organisms cannot grow on agar medium [210], making their isolation more challenging, as their optimal growth temperatures exceed the melting point of agar [211]. More importantly, as the majority of rumen micro-organisms are anaerobic, creating and maintaining an anaerobic environment in the laboratory is often challenging and costly. For instance, anaerobic chambers require continuous circulation of non-inert gas mixtures (e.g. N₂ and CO₂) and real-time monitoring, as well as periodic and expensive maintenance. Particularly for the cultivation of methanogenic archaea, high concentrations of hydrogen gas are often necessary, which poses significant safety risks due to its flammability [212].

Certain small signalling molecules, such as the resuscitation-promoting factor [213], 5-amino-acid peptide [214] and cell wall muropeptides [215], can be used to resuscitate the viable but non-culturable bacteria in dormancy [216,217]. However, there is no universal method for resuscitating all dormant bacteria due to their immense diversity and the potential evolution of novel dormancy mechanisms over time. Additionally, some species have symbiotic interdependencies, and thrive only by sharing and exchanging molecules or electrons with other species [218,219], and often require physical contact or spatial proximity, further complicating their separation. According to Kenter's hypothesis [206], some of these bacteria may facilitate the growth of their symbiotic counterparts but are incapable of independent growth. In some cases, however, abiotic substances were used successfully as substitutes for the symbiotic partners during isolation, offering a potential solution to this challenge [220].

Finally, within the finite substrates, the competition between microbes may result in the disadvantaged position of low-abundance or low-substrate-affinity species [221]. In liquid culture systems, highly adaptable microbial species tend to dominate the community and can be isolated repeatedly, as demonstrated by the dominance of streptococci in the study by Rute et al. [205]. The dilution-to-extinction method has proven effective in isolating low-abundance unculturable bacteria [222] through its capacity to reduce the competitive advantage of dominant species [206,207], thereby facilitating the cultivation of previously uncultured micro-organisms. Notwithstanding, it was shown by Mao et al. that certain rare species seem to be more sensitive to the dilution level as their richness was decreased further compared with others [223]. These complexities suggests that we are currently unable to isolate all micro-organisms within communities with a general method. In other words, targeting a particular taxa during isolation and culturing will inevitably lead to the sacrifice of other taxa or portions of the population. However, despite these challenges, the emerging field of culturomics holds promising perspectives for future advancements.

4.2. Innovative solutions for mitigating the challenges of unravelling the risks of AMR in the rumenofaecobiome

4.2.1. Standardization of anaerobic culturing and testing methods

A key challenge in studying the rumenofaecobiome is maintaining strict anaerobic conditions throughout culturing and testing, as even brief oxygen exposure can compromise bacterial viability and affect results [119,224]. To address this, access to standardized anaerobic culturing infrastructure and protocols must be established. Preserving anaerobic microbial diversity is also critical. Huws et al. [225] recommend that journal publishers should require isolate storage in culture collections before publication to prevent the loss of valuable anaerobic strains, which can become irrecoverable over time as culture collections offer better infrastructure for preserving and sharing strains, facilitating open re-

search. Another challenge is the slow growth of many anaerobic bacteria, making conventional AST methods and clinical breakpoints developed for aerobes unsuitable. Establishing authoritative research centres focused on anaerobic AMR studies can accelerate method development and ensure appropriate AST protocols for anaerobes in general, and for rumenofaecobiome isolates in particular. This could also include the development of automated AST systems to minimize human error and variability, as well as the use of PCR-based ARG detection techniques [226] to complement traditional culturing, providing faster results and improving diagnostic accuracy. It is proposed that by integrating standardized culturing and testing protocols, researchers can generate more reliable AMR data, enabling effective antibiotic stewardship strategies. This will help to mitigate the spread of AMR while promoting sustainable livestock production.

4.2.2. Development of novel culture media

Hungate (1966) pioneered the use of clarified rumen fluid in culture media to better replicate the native microbial environment [227]. More recently, Rute et al. [205] demonstrated that culture media composition significantly affects the bacterial communities cultivated from rumen samples. Boca Raton [228] and Kenters [206] media produced microbial communities well established to be part of the rumen environment, whereas Nyonyo [229] medium facilitated the isolation of a higher proportion of bacteria in unclassified genera, increasing the likelihood of recovering previously uncultured species. This highlights the need for continued development of specialized culture media to enhance the recovery of anaerobic rumen micro-organisms.

Metabolomics offers a powerful approach to optimizing culture conditions by identifying growth factor requirements and metabolic pathways of uncultured micro-organisms [230]. Metagenomic data can be utilized to analyse metabolic pathways among as-yet-unculturable microbial communities [123]. By comparing gene sequences and referencing metabolic pathway databases such as KEGG [231], researchers can reconstruct metabolic pathways, identify key functional genes, and determine how novel micro-organisms utilize carbon, nitrogen and other nutrients. This information can provide a scientific basis for designing selective media tailored to unculturable taxa.

To enhance selectivity, metabolic analysis results can be integrated with AMR profiles, allowing the inclusion of targeted antimicrobials to suppress non-target microbial communities. Shaffer et al. introduced the Distilled and Refined Annotation of Metabolism model [232], which automates the conversion of genomic data into metabolic characteristics, significantly reducing the time and effort required for metabolic analysis. This tool or similar tools could support the rapid identification of metabolic traits in uncultured micro-organisms, and accelerate the development of optimized culture media that are needed to advance rumenofaecobiome studies.

4.2.3. Advancing gene assembly, prediction and annotation: development of high-efficiency computational tools

To address the computational and time constraints associated with large-scale functional profiling of metagenomes, and to better understand the impact of mistranslated DNA sequences in the interpretation of biological traits, Dimonaco proposed a novel model called 'FrameRate' [145]. This model leverages machine learning to predict coding frames directly from unassembled DNA sequencing reads, eliminating the need for homology-based inference or pre-computed databases, which significantly reduces the computational burden and enhances the efficiency of metagenomic analysis.

On the other hand, for gene annotation, existing tools are not universally applicable to all genome studies and share a common

limitation: a significant portion of genomic regions, particularly intergenic regions, remain unannotated. Even in extensively studied model organisms such as *E. coli*, a significant proportion (35%) of genes remain functionally uncharacterized and are classified as hypothetical [233]. To tackle this issue, tools such as ORForise and StORF-Reporter [145] can be employed for the re-annotation and identification of previously unknown CDSs missed by the traditional annotation methods, addressing the omissions of pseudogenized or functional genes due to in-frame stop codons or the alternative use of stop codons to encode amino acids. The development of artificial-intelligence-driven resistome analysis tools is also critical.

4.2.4. Customization of rumen-associated AMR databases

The rumen and ruminant faecal environments are highly complex and unique [42,234], hosting a diverse array of micro-organisms whose genomes and AMR profiles differ significantly from those found in humans or other environments [6]. General AMR databases, typically built from clinical or environmental samples, may not sufficiently capture the distinctive ARGs and microbial diversity present in the rumen-relevant environments [235]. Therefore, developing a bespoke/custom rumenofaecobiome-associated AMR database is crucial for accurately identifying and annotating ARGs in this environment, which would facilitate the study of AMR evolution and adaptation in these environments while providing better understanding of transmission dynamics. Such a bespoke database could not only enhance our understanding of AMR mechanisms within the rumenofaecobiome, but also provide valuable insights for optimizing antibiotic use in livestock. For example, in other areas of rumen studies, Wilkinson et al. [236] developed a dedicated version of the Phylogenetic Investigation of Communities by Reconstruction of Unobserved States tool, named 'CowPI', designed specifically for rumen microbiome research. The functional profiles predicted by CowPI aligned more closely with estimates derived from metagenomic and metatranscriptomic datasets of the rumen, being able to capture higher genetic variation and larger pangenomes.

4.2.5. Reducing antimicrobial usage and development of new antibiotics and alternatives

Reducing antimicrobial usage in global livestock farming is an avenue to help reduce the emergence of AMR and dissemination between humans, animals and the environment [237]. However, the consumption of antimicrobials is inevitable in our daily lives due to the high demand for infection control and treatment. Moreover, the development of AMR is the irreversible natural evolution outcome of using antimicrobials. To mitigate the indiscriminate usage of antimicrobials in agriculture and health care, it is essential to not only discover more effective antimicrobials, but also actively pursue the development of effective alternatives for preventing or treating related diseases. This necessitates a comprehensive understanding of both the mechanisms of antibiotic action and the processes underlying resistance development, providing a foundation for the design of novel therapeutic agents. Indeed, research exploring alternatives to antimicrobials, especially from underexplored sources including from the rumen, is on the increase. For example, Shukla et al. discovered clovibactin, a novel antibiotic from an unculturable soil bacteria which was effective against resistant pathogens [126]. In-silico analyses and culture-independent functional metagenomics have identified several rumen-derived antimicrobial peptides (AMPs) with potential against clinically important pathogens in veterinary and human medicine [154,238–242]. Due to their more diverse mechanisms of action and broader range of targets, AMPs are less likely to induce drug-resistant mutations in micro-organisms compared with conventional antimicrobials [243].

They have the potential to treat bacterial and viral infections, inhibit the transmission pathways of AMR, enhance immune pathways, and improve livestock production [244–246]. It is proposed that the use of these low-resistance-inducing compounds for infection control in ruminants will help to reduce ARG development and acquisition in microbiome populations, thereby aiding the complexities of AMR transmission in these environments to be unravelled. This is a worthy line of intensified enquiry for future research comparing conventional antimicrobials with these newer alternatives in their ability to induce AMR within ruminant populations.

Similarly, the use of plant-based compounds (e.g. citrus flavonoid extract) has been shown to maintain dairy cow health and rumen stability by regulating AMR levels in the microbiota [155]. Bacteriophage therapy is also being considered as a treatment strategy for combating AMR infections in humans, plants and animals [156,247]. Additionally, some researchers have proposed that reconstructing a healthy gut microbiota to enhance its colonization could effectively mitigate the negative impacts caused by drug-resistant bacteria [248]. Moreover, the integration of computational technologies in biotechnology (e.g. the application of artificial intelligence [157] for breakthrough drug development and the use of machine-learning assemblers which simulate the flow network of MGEs carrying ARGs in different environments [249]) are interdisciplinary technologies that can help track and block the spread of ARGs.

5. Studying the rumenofaecobiome: long-term implications for policy and global AMR management

From a biomedical perspective, studying the proposed rumenofaecobiome presents many possibilities. The as-yet-unculturable micro-organisms in the rumenofaecobiome represent a valuable yet largely underexplored resource for discovering new ARGs and mechanisms, as well as new antimicrobials and other bioactive compounds. These novel compounds, along with newly discovered organisms and their enzymes, could contribute to the development of innovative strategies to mitigate AMR in both human and veterinary medicine. For instance, certain organisms or their enzymes may act as AMR blockers in the ruminant gut [250], offering potential as feed additives or live feed technologies to combat AMR [251]. Additionally, as-yet-unculturable probiotics can be used to enhance feed efficiency in livestock while preventing and mitigating the incidence of common diseases [19,23,251]. This not only generates economic benefits for the livestock industry by reducing operational costs, but also decreases expenditure associated with veterinary healthcare maintenance.

The new microbial communities identified through a consolidated approach, such as rumenofaecobiome analysis, could also play a crucial role in advancing our understanding of the mechanisms of antimicrobial action and resistance within the ruminant gut microbiome. This includes uncovering metabolic pathways linked to current management practices that may inadvertently aggravate AMR acquisition and spread [252]. By gaining deeper insights into these processes, more precise treatment regimens can be designed, reducing reliance on broad-spectrum antibiotics and, consequently, the risk of the spread of AMR.

From a broader political and economic perspective [253], improved livestock health enhances the safety and efficiency of agricultural and food supply chains, boosting the development potential of agricultural and livestock products in international trade. Moreover, by improving our understanding of AMR dynamics in the rumenofaecobiome, we can curb the spread and exchange of AMR within the One Health framework (i.e. to humans and the environment), and thus public health risks are reduced significantly, leading to lower healthcare costs and enhanced societal well-being.

Furthermore, the discovery of economically and biotechnologically important enzymes from the microbial communities within the rumenofaecobiome could open new avenues for industrial applications, such as biofuel production [120], waste management [254] and the development of novel bioproducts [255]. These advancements could contribute collectively to sustainable development, environmental health and economic growth.

The untapped potential of uncultured microbes within the rumenofaecobiome is also likely to attract increased investment in other areas, and collaboration between biology and computational science could further drive innovation in the industry, fostering global economic growth. For example, artificial-intelligence-assisted AMR management systems can rapidly process and interpret data, enabling healthcare and veterinary teams to adapt quickly to changing conditions and prevent unnecessary antimicrobial prescription [158]. This multi-disciplinary collaboration aligns with the One Health concept, which highlights the interconnectedness of human, animal and environmental health. By integrating these discoveries into sustainable management practices, we can create a holistic approach to combat AMR of ruminant livestock origin while promoting economic and environmental sustainability.

6. Conclusions

The exploration of AMR within rumen bacteria by studying the rumenofaecobiome has critical implications for the One Health initiative. This review highlights the intricate interactions within microbial communities in ruminant livestock and their broader environmental and public health connections. Studying the rumenofaecobiome in ruminants will provide insights into the overall microbial ecosystem, its role in AMR transmission, and the potential impact on both animal and public health. This approach is valuable as the faecal microbiome is often used as a proxy for the rumen microbiome in resistome studies, despite certain limitations in representing as-yet-unculturable rumen bacteria. Furthermore, research into both culturable and as-yet-unculturable ruminal bacteria will provide essential insights into the mechanics of ruminal ARG transmission and resistance element mobilization in the One Health framework.

While significant strides have been made in understanding the microbiomes of livestock, challenges persist in fully characterizing the resistome of ruminants due to their complex digestive systems and difficulties in culturing their diverse microbial populations. Continued advancements in genomic technologies and bioinformatics approaches, such as metagenomics and spatial transcriptomics and culturomics, are essential for unravelling these complexities, understanding ARG dynamics, and mitigating the spread of AMR in, from and to livestock, thereby protecting livestock, environmental and public health. The solutions proposed here offer a comprehensive view of potential approaches to understanding resistance mechanisms in this unique environment, which is essential for developing effective AMR mitigation strategies. Integrating comprehensive databases, analytical tools and new automated *in-vitro* AST technologies, standardization of anaerobic ASTs, development of new media, funding of anaerobic infrastructure, and training of personnel are all instrumental in dissecting the complex genetic frameworks underlying AMR, especially in the unculturable populations of the rumenofaecobiome, and for aiding precise interventions with far-reaching effects. Innovations in genomic technologies and global data-sharing of genetic resources are also pivotal for enabling the detailed mapping of resistance pathways, and fostering international collaboration in the ongoing battle against AMR of ruminant livestock origin.

This review underscores the need for a transdisciplinary strategy to address AMR, combining genetic, phenotypic and ecological research to forge effective control measures. To safeguard One

Health, it is imperative to continue advancing these innovations for improved diagnostic capabilities to enhance global surveillance and response systems against the ever-evolving threat of AMR and the related economic impacts.

Author contributions

ZW, FS, SH and LO conceived the project. ZW, MF, TS, FB, GD, FS, SH and LO contributed to the study design. ZW designed the layout and figures. ZW and LO wrote the manuscript with input from all authors, including MF, TS, FB, GD, FS, SH and LO. All authors contributed to proofreading the manuscript.

Funding: This research was supported by funding from the Biotechnology and Biological Sciences Research Council, UK (Ref. BB/X012794/1).

Competing interests: None declared.

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